

BLink Bead Preparation Protocol

Overview:

The BLink bead technology is based on magnetic, barcoded beads loaded with Tn5 protein to generate active transposomes. The preparation of these beads requires a series of barcode ligations (split-pool format) to build out the final combinatorial barcode plus Tn5 mosaic end (ME) sequence. Each bead has many copies of the same barcode; total barcode complexity in a batch of beads is > 7M combinatorial barcodes. A ‘staggered’ barcode segment of variable length ensures the sequencing reads are high complexity, even through the short invariant regions of the barcoded region. The barcode is captured in R1 on Illumina sequencers. See separate “BLink-seq Library Prep Protocol” to use BLink beads to generate linked-read libraries.

The preparation of a batch of beads takes ~1 week to complete, ideally with 2-3 people to maximize efficiency of the split-pooling steps.

- Day 1** Recommended: prepare stock buffers and duplex-oligo stocks, if needed
Bind biotinylated-duplex oligo to beads (bulk) – overnight incubation
- Day 2-3⁺** Split beads to one 8-well strip, ligate Stagger-duplex, wash, pool
Split beads to 12 x 8-well strips, ligate Segment1-duplex (96 barcodes), wash, pool
Split beads to 12 x 8-well strips, ligate Segment2-duplex (96 barcodes), wash, pool
Split beads to 12 x 8-well strips, ligate Segment3-duplex (96 barcodes), wash
Ligate ME-duplex, wash, final pooling
- Day 3⁺** Test the bead batch:
Strip one aliquot of beads with NaOH to remove duplex oligos and anneal the ME-REV oligo
Test-load small amounts of beads with a Tn5 dilution series
Prepare haplotagging libraries with a control DNA sample to determine the optimal Tn5-loading dilution
- As needed** Strip aliquots of beads with NaOH and anneal the ME-REV oligo
Load beads with Tn5 protein
Recommended: pool all beads after stripping and/or loading large batches, and store as a homogeneous stock.

BLink beads can be stored in several formats (unstripped, stripped+annealed to ME-REV, loaded with Tn5) at 4°C, in the appropriate buffer as described in the protocol. Beads can be stripped/annealed and loaded with Tn5 as needed to prepare the final transposome-active haplotagging bead stock, as the best long-term storage may be in the unstripped format. The loading seems to be consistent for beads stripped and/or loaded at different times, though test loading can be repeated as needed to confirm the appropriate Tn5 loading concentration.

Materials and Equipment Required: see Appendix A and BLinkBead_oligos.xlsx

Duplex-oligo stocks: see Appendix B to prepare duplex-oligo stocks used in ligation reactions

Technology development: Jennifer Grenier¹ designed BLink beads, modified from the original ‘haplotagging beads’ designed by Frank Chan² and Marek Kucka². All of the authors of the manuscript “The authors of the manuscript “BLink-seq delivers population-scale haplotypes without long reads: a scalable framework for non-model genomics” contributed to developing BLink-seq.

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Section 1: Split-pool ligations to generate combinatorial barcodes

Binding of 2xBiotinylated oligo duplex to beads

Day 1 mid-afternoon

1. Prepare 25 μ M 2xBiotin-TS5P oligo-duplex stock, if needed (see Appendix B).
2. Mix the M-280 streptavidin bead bottle well to ensure beads are fully resuspended.
3. Transfer 3mL of M-280 streptavidin beads to a 5mL tube and place on a magnetic stand.
4. Remove storage buffer and add 5ml of streptavidin binding buffer with Triton (1x SBB+T). Mix well off-magnet, return to magnet. Wait for the beads to pellet and the buffer to clear.
5. Remove the buffer and replace with 4 mL 1x SBB+T. Remove from magnet and mix well to resuspend the beads.
6. In a separate tube, combine 1 mL 1x SBB+T with 5 μ l 25 μ M 2xBiotin-TS5P oligo-duplex. Mix well by gently vortexing.
7. Transfer 1 mL 2xBiotin-TS5P oligo-duplex diluted in 1x SBB+T to the beads (final concentration 0.025 μ M). Immediately close the tube and invert to mix well.
8. Place the beads on a rotator for 30 minutes at room temperature (RT), and then rotate overnight at 4°C.

Day 2 morning

Prepare for split-pool ligations by

- preparing oligo-duplex stocks, if needed (see Appendix B), or thawing stocks on ice
 - preparing 1x WASH buffer+biotin (50mL minimum, prepare more as needed)
 - bringing the beads to room temperature on the rotator.
9. Pulse-spin the beads, place on magnet, and remove the buffer when clear.
 10. Add 5 mL of WASH+biotin buffer. Mix by inverting, rotate 10 min at RT.
 11. Return to magnet, wait to clear, and remove the wash. Take care with any liquid in the lid.
 12. Repeat the wash with another 5 mL of WASH+biotin buffer for 10 min at RT.
 13. Return to magnet, wait to clear, and remove the wash. Take care with any liquid in the lid.
 14. Add 500 μ L WASH+biotin buffer; close and flick the tube to fully disperse the beads.

Split-pool ligation: Stagger oligos

15. Prepare Stagger-Ligation MM for 1 strip (8 reactions, 35 μ l per reaction plus 5% excess). Hold on ice.
This ligation reaction is 2.4x more concentrated than 96x barcode-ligations.

149 μ l 2x StickTogether buffer
128 μ l nuclease-free (NF) water
<u>21 μl T7 ligase</u>
298 μ l total MM in 1x StickTogether buffer

16. In a PCR strip tube, dilute 8 stagger-duplex oligos to 2.5 μ M in 1x StickTogether buffer. Hold on ice.

15 μ l 2x StickTogether buffer
12 μ l NF water
<u>3 μl 25 μM stagger-duplex stock</u>
30 μ l total for each of 8 stagger-duplex oligos

Split beads into a PCR strip:

17. Aliquot 50 μ l of beads to one 8-well PCR strip.
18. Bring the volume of the leftover beads to over 400 μ L with WASH+biotin and aliquot 50 μ l into each well in the strip.
19. Bring the volume of the leftover beads to 50 μ L with WASH+biotin and aliquot 10 μ l into each well.
At this point, there is ~110 μ l volume (containing beads) in each well in the strip.
20. Place the strip on a plate magnet; wait to clear.

Ligate the stagger-duplexes:

21. On magnet, remove all WASH+biotin buffer from the strip.
22. Remove strip from magnet. With a multichannel pipet, add 25 μ l of the diluted stagger-duplex oligo stocks from step 17 to the strip. Immediately close the lids and flick to mix until beads are fully re-suspended.
Briefly pulse-spin if needed to collect the liquid at the bottom of the tube, but the beads should stay fully re-suspended.
23. Add 35 μ l Stagger-Ligation MM to each tube, then immediately close and flick to mix. *Briefly pulse-spin if needed.*
24. Incubate in the Thermomixer for at least 15 min at 24°C, shaking at 800 RPM.
25. After 15 min ligation, pulse-spin the strip and place on a plate magnet.
26. Remove the supernatant, leaving the strip on the magnet.
27. Add 200 μ l of WASH+biotin per well on-magnet, pipet up and down to rinse, then remove all the wash.
28. Add 150 μ l of WASH+biotin buffer per well (optional: re-rack tips). Close strips, invert to mix, rotate 10 min at RT.
29. Pulse-spin, place on magnet and remove wash when clear (optional: using re-racked tips).
30. Repeat the washing step with another 150 μ l WASH+biotin buffer (optional: re-racking tips), rotate 10 min at RT.
31. Pulse-spin, place on magnet and remove wash when clear (optional: using re-racked tips).
32. Add 50 μ l of WASH+biotin buffer per well.

Pool all beads back to one tube:

33. Transfer all beads from the strip into a new 5 ml tube containing 3 mL WASH+biotin buffer, and place on magnet.
Rinse the strip with more WASH+biotin buffer and transfer to the 5 ml tube to recover all of the beads.
34. Remove buffer, wash again with 5mL WASH+biotin buffer.

Split-pool ligation: barcode segment 1

35. Prepare Ligation MM for 12 strips (96 reactions, 7 μ l per rxn, 108x with excess):

378 μ l 2x StickTogether buffer
324 μ l NF water
<u>54 μl T7 ligase</u>
754 μ l total

36. Divide the Ligation MM into 8-well strip reservoir, ~90 μ l per well. Hold on ice.
37. Dilute the Segment1 barcode-duplex stocks to 1 μ M in 1x StickTogether buffer in a new 96 well plate:
 - Combine 780 μ l 2x StickTogether and 620 μ l NF water in a 1.5ml tube and mix well.
 - Using a strip-reservoir and a multichannel pipet, distribute 9 μ l to each well in a 96 well plate.
 - Transfer 1 μ l of the 10 μ M segment 1 barcode-duplex oligo stocks to each well. Hold on ice.

The diluted stocks can be mixed by pipetting at the transfer step, or the plate can be sealed and vortexed/pulsed down.

Split beads evenly into PCR strips:

The goal is to evenly distribute the beads into 96 tubes (12 x 8-well strips) with a multichannel pipet, using a PCR strip as a reservoir. Beads need to stay well mixed/resuspended during this process.

38. Number 12 PCR-strip tubes for the first barcode-ligation reaction. Each strip needs a number, but not each well.
39. Mix the beads well (in 5mL WASH+biotin buffer), and transfer ~200 μ L x 8 into a PCR strip tube reservoir.
40. With a multichannel pipet, transfer 50 μ L of beads to each numbered strip. Mix and refill the strip-reservoir as needed.
41. Bring the volume of the leftover beads to 5mL with WASH+biotin. Mix well and transfer to the strip-reservoir.
42. With a multichannel pipet, transfer another 50 μ L of beads to each strip. Mix and refill the strip-reservoir as needed.
43. Bring the volume of the leftover beads to 1mL with WASH+biotin and transfer to the strip-reservoir.
44. With a multichannel pipet, transfer another ~10 μ L of beads to each strip. Mix and refill the strip-reservoir as needed.
At this point, the 12 strip tubes contain ~110 μ L volume (with beads) in each well.
45. Place strips on plate magnet.

Add segment1-duplexes (in 1x StickTogether), one strip at a time:

46. On magnet, remove all WASH+biotin buffer from one strip (optional: reuse tips for multiple strips, they are identical).
47. Remove strip from magnet. With a multichannel pipet, add 5 μ L of 1 μ M segment1-duplex (prepared in step 37) to the beads, matching the column with the strip number. Close and flick (or invert over a bar magnet) until the beads are fully re-suspended.
48. Repeat for each strip / segment1-duplex column, leaving the strips off-magnet at RT. Do not continue until all segment1-duplexes have been added and each strip is mixed well.

Add Ligation MM, one strip at a time:

49. With a multichannel pipet, add 7 μ L Ligation MM to each well in strip 1, then immediately close and flick to mix.
Briefly pulse-spin to collect most liquid at the bottom but not pellet the beads, and leave the strip off-magnet.
50. Proceed with adding 7 μ L Ligation MM per well to the other strips, mixing each strip immediately.
51. Incubate in the Thermomixer for 15 min at 24°C, 800 RPM.
52. After 15 min ligation, pulse-spin and place the strips on magnet.
53. Quick rinse on-magnet: add 100 μ L of WASH+biotin per well, then immediately remove all of the wash.
54. Slow wash: add 150 μ L of WASH+biotin buffer per well (re-racking tips).
Close strips, invert to mix, rotate 10 min at RT.
Pulse-spin, place on magnet and remove wash when clear (using the same tips used to add the wash for each strip).
55. Repeat the slow wash step with another 150 μ L WASH+biotin buffer (re-racking tips).
Close strips, invert to mix, rotate 10 min at RT.
Pulse-spin, place on magnet and remove wash when clear (using the same tips used to add the wash for each strip).
56. Add 50 μ L of WASH+biotin buffer to each strip.

Pool all beads back to one tube:

57. Consolidate all beads to one strip using a multichannel with one set of tips as follows:
 - Place one strip on the magnet – *all of the beads will consolidate to this strip.*
 - Resuspend the beads in another strip and transfer to the strip on the magnet.
 - Use the buffer in the magnet-strip to back-rinse the empty strip as needed, to collect all of the beads on-magnet.
 - Repeat with each additional strip, removing the buffer in the strip on the magnet when it fills up (~4 strips = 200 μ L).

58. Transfer all beads from the consolidated strip into a new 5 ml tube containing 3 mL WASH+biotin buffer, and place on magnet. Wash the strip with more WASH+biotin buffer and transfer to the 5 ml tube.
59. Remove buffer, wash again with 5mL WASH+biotin buffer.

Repeat steps 38-59 for two more rounds of split-pool ligations for barcode segments 2 and 3

The full process of ligating the stagger duplexes and all 3 barcode segments typically requires 2 days. The best time to stop is in the second wash step after the ligation reaction or after pooling all of the beads (rotate overnight at 4°C).

Briefly:

- Setup: Prepare more of the Ligation MM
Dilute the next plate of barcode segment-duplex oligos in 1x StickTogether buffer
Evenly divide the bead stock to 96 wells in new numbered strips and place on magnet
- Remove buffer from beads, add 5 µl 1 µM segment2 or segment3 duplex oligos.
- One strip at a time, add 7 µl Ligation MM, close/mix/quick pulse.
- Incubate 15 min at 24°C, 800 RPM.
- Pulse-spin, place on magnet, when clear do a quick rinse with 100 µl of WASH+biotin buffer
- Two slow washes with 150 µl of WASH+biotin buffer (re-racking tips) – stop here after ligating segment3!
- Final 50 µl of WASH+biotin buffer, then repool all beads using the consolidation method
- Pool all beads from the consolidated strip into a new 5 ml tube containing 3 mL WASH+biotin buffer
- One more wash with 5 mL WASH+biotin buffer

Ligate Ph-shortME/MErev-LIG (TGG overhang)

60. Prepare ME-duplex Ligation MM for 12 strips (96 reactions, 12 µl per rxn): 108x recipe

648 µl 2x StickTogether buffer
513 µl NF water
54 µl T7 ligase
1 µl 10 µM duplex Ph-shortME/MErev-LIG duplex
1.296 mL total

61. Divide into 8-well strip reservoir, ~160 µl per well. Hold on ice.
62. Place strips on magnet and remove all WASH+biotin buffer (OK to reuse tips).
63. One strip at a time, add 12 µl ME duplex ligation MM to each well, immediately close and invert to mix.
64. Incubate in the Thermomixer for 15 min at 24°C, 800 RPM.
65. After 15 min ligation, pulse-spin and place the strips on magnet.
66. Add 100 µl of WASH+biotin per well on-magnet (quick-rinse), then remove all the wash from the strips.
67. Add 50 µl of WASH+biotin buffer per well.
68. Pool all beads using the consolidation method.
69. Transfer all beads from the consolidated strip into a new 5 ml tube containing 3 mL WASH+biotin buffer, and place on magnet. Wash the strip with more WASH+biotin buffer and transfer to the 5 ml tube to collect all of the beads.
70. Remove buffer, add 5mL WASH+biotin buffer. Remove from magnet.

Safe stop: store at 4°C (do not freeze beads!).

Section 2: Strip duplex oligos from beads and anneal ME-REV oligo

At this point, BLink beads have completed the split-pool ligation steps but still have the full barcode-duplex oligos. The reverse-strand oligos do not get ligated to each other because there is no 5' phosphate, and are not attached to the beads other than by the duplex hybridization. These oligos are removed with 2 rounds of NaOH denaturation and wash steps. Then the ME-REV oligo is annealed to the bead stock prior to the final step of loading Tn5 protein.

In the standard stripping protocol below, 415µl of the ligated bead stock is stripped with 1mL 0.15M NaOH. This can be scaled down if desired, or scaled up by processing multiple tubes of 415µl of the ligated beads in parallel.

These steps require 2 Thermomixers with 1.5mL blocks, one set to 55°C and a second set to 40°C. If only 1 Thermomixer is available, remove the block from the base immediately after the final incubation at 55°C to promote rapid cooling to 40°C.

1. Prepare buffers: volumes indicated are sufficient to strip 1 tube of 415µl of ligated bead stock, and can be scaled as needed for a smaller volume of beads or for multiple tubes stripped in parallel.
 - Prepare 2.2 ml NaOH stripping buffer. Hold at room temperature.
 - Prepare 6 ml WASH+T buffer.
 - Transfer 1.2mL WASH+T buffer to a new tube and pre-warm in a Thermomixer at 55°C.
Use the heated lid if available.
 - Transfer 1.1 mL WASH+T buffer to another tube and add 11µl 100 µM ME-REV oligo. Hold on ice.
The final concentration is 1 µM ME-REV oligo.
 - Save the remaining WASH+T buffer at room temperature.
2. Mix the ligated bead stock by inversion or rotation until fully dispersed in suspension.
3. Transfer 415µl to new 1mL tubes and place on magnet.
The timing of the stripping steps should not be extended: if processing more than 4 tubes in parallel, consider processing in smaller batches to avoid excessive incubation in NaOH.
4. Remove all WASH+T buffer from the beads and remove from magnet.
5. Add 1 mL NaOH stripping buffer. Immediately close the tube and mix by inverting several times until no bead pellet is visible. Incubate the tube on a rotator for max 1 min at room temperature, then pulse spin and place on magnetic rack (the beads should be in contact with NaOH for at most 2 minutes).
6. On magnet, as soon as the beads are pelleted, remove all NaOH stripping buffer (save the waste for appropriate disposal).
7. Remove the tube from the magnetic rack and immediately add 1.2 mL **warm** WASH+T buffer. Mix by inverting until no bead pellet is visible.
8. Rotate beads in WASH+T buffer for 1 min at room temperature; pulse-spin and return to magnet.
9. While beads are washing:
 - Refill the pre-warmed WASH+T tube to 1.2mL and pre-warm on the Thermomixer at 55°C.
 - Pre-warm the tube of WASH+T buffer containing 1 µM ME-REV to 55°C in the Thermomixer.
10. Repeat NaOH stripping steps:
 - On magnet, remove WASH+T buffer from the beads
 - Add 1mL NaOH stripping buffer, mix, rotate max 1min at room temperature, pulse spin, and return to magnet.
 - As soon as beads have pelleted, remove NaOH stripping buffer (save the waste for appropriate disposal)
 - Add 1.2mL **warm** WASH+T, rotate 1 min at room temperature, pulse and return to magnet.
11. When beads have pelleted, remove all WASH+T buffer and add 1 mL **warm** WASH+T buffer with 1 µM ME-REV.
12. Rotate the tube for 5 min at room temperature.
13. Transfer the tube to a Thermomixer set to 40°C for 5 min at 800 RPM.

14. Return to the rotator at room temperature and continue to mix for another 15 min.
15. Prepare WASH+T+Biotin buffer: in a new tube, combine 1.4mL WASH+T buffer and 3.5uL 200mM Biotin. Vortex to mix; hold at room temperature.
16. After the 15 min incubation, pulse-spin the tube containing the annealed beads and place on a magnet.
17. When beads have pelleted, remove the WASH+T+ME-REV buffer.
18. Add fresh 1.2 mL Wash+T+Biotin buffer and rotate for another 20 min at room temperature.

The tube now contains beads ready for Tn5 loading.

Store the stripped beads annealed with ME-REV in WASH+T+Biotin buffer at 4°C.

Section 3: Tn5 loading

Small scale Tn5 loading to find the optimal Tn5 concentration

Goal: Titrate amount of Tn5 protein to achieve optimal tagmentation of gDNA to approximately 450-1200bp library size.

At this point, BLink beads have been annealed with ME-REV and are ready to load Tn5. The protocol describes loading 4 different Tn5 concentrations, but can be scaled to as many test loads as desired. Typically, 25µl of the ME-REV annealed bead stock is loaded in a final volume of 50 µl.

ME-B/ME-REV duplexed oligos (ME-B duplex) can also be added during the bead load. While this ME-B is not sufficient to generate the appropriate insert sizes for Illumina sequencing, it may help to make tagmentation more efficient on the bead surface. ME-B transposome is supplemented during library prep as well. See Appendix B to prepare 50 µM ME-B/ME-REV duplexed oligo stock; make additional dilutions in 1x Annealing buffer after duplex formation.

Diagenode unloaded Tn5 protein is very concentrated (approx. 100µM), and can be diluted to prepare a working stock in 30% or 50% glycerol storage buffer. If preparing excess diluted unloaded Tn5 protein stock, dilute in 50% glycerol storage buffer and store the remainder at -20 °C. The typical loading amount is ~8nM Diagenode unloaded Tn5 protein, or 4µl of a 1:1000 dilution of the primary unloaded Tn5 stock per 50µl bead load.

Example for 4 test loads:

Test load #				
1	2	3	4	Bead loading reagents
46	45	44	43	µl 30% glycerol storage buffer
1	1	1	1	µl 0.2µM /ME-REV duplex oligos (optional)
3	4	5	6	µl unloaded Tn5 protein (1:1000 dilution)

1. Mix the stock of BLink beads annealed with ME-REV on a rotator until fully dispersed.
It can help to lift the beads out of the bottom of the tube on a magnetic rack before placing on the rotator.
2. Transfer 100µl to a new 1.5 ml tube (*sufficient for 4 tests using 25µl orig beads/test*) on a magnetic stand.
3. When the beads have pelleted on the magnet and the WASH+T buffer is clear, remove the WASH+T buffer.
4. Remove the tube from the magnetic rack and add 105µl of WASH+T buffer. Mix well.
5. Aliquot 25µl into 4 wells of a new PCR strip and place on a magnetic stand.
6. When the beads have pelleted on the magnet and the WASH+T buffer is clear, remove the WASH+T buffer.
7. Remove the strip from the magnetic rack and add 30% glycerol storage buffer (volume based on the test loading plan). Close the strip and flick to disperse the beads.
8. Optional: add ME-B duplex oligos to each tube, final concentration 4nM. Close the strip and flick to mix.
9. Add unloaded Tn5 protein (diluted) to bring volume to 50 µl. Immediately close the strip and flick to mix well.
10. Rotate at least 10 minutes at room temperature, then overnight (or longer) at ~4°C. Beads may continue to load over several days.

Store Tn-5 loaded BLink beads at 4°C.

11. Follow the 'BLink-seq: library prep protocol' with a control gDNA sample to test Tn5 loading, using 5µl of the Tn5-test loaded beads per library. It is also important to titrate the ME-B transposome (in solution) to optimize yield and fragment size distribution, e.g. prepare libraries with 2 or more ME-B dilutions per bead test load. Typically, bead performance will plateau and adding more Tn5 will not alter library yield or size distribution above the optimal loading concentration.

Batch loading of Tn5

After determining the optimal amount of Tn5 to use per 50µl bead load, the loading protocol can be scaled up as desired. Maintain the same Tn5:bead ratio, scaling the volume up to max 500µl loaded beads per 1.5mL tube. The final volume of loaded beads will be double the starting volume of annealed, unloaded beads.

For example:

1. Mix the stock of BLink beads annealed with ME-REV on a rotator until fully dispersed.
It can help to lift the beads out of the bottom of the tube on a magnetic rack before placing on the rotator.
2. Transfer the desired volume of beads to one or more 1.5mL tube (max 250µl annealed bead stock per tube, to load in max 500µl 30% glycerol storage buffer).
3. Place on a magnetic rack. When the beads have pelleted on the magnet and the WASH+T buffer is clear, remove the WASH+T buffer.
4. Remove the tubes from the magnetic rack and add 500µl fresh WASH+T per tube. Mix well to disperse the beads.
5. Pulse spin and place the tubes back on the magnet.
6. Prepare a loading master mix containing 30% glycerol storage buffer, ME-B duplex (optional), and unloaded Tn5 protein, with total volume sufficient for the batch load.
7. Remove the WASH+T from the beads and add up to 500µl loading master mix. Immediately mix to disperse the beads.
8. Place the tube(s) on rotator at room temperature for minimum 10 minutes, then rotate overnight or longer at 4°C.
The loading performance appears to improve after incubating at 4°C for several days.
9. Pulse spin down the tubes. If several tubes are loaded per batch, they can be mixed together to ensure the batch is homogeneous.

Store Tn-5 loaded BLink beads at 4°C.

10. Recommended: run final library prep tests to confirm the batch is performing as expected. The amount of ME-B transposome required during library prep may need to be optimized for each batch.

Appendix A: Materials and Equipment Required

Reagents and Supplies:

Dynabeads™ M-280 Streptavidin	ThermoFisher #11206D
T7 ligase (with 2x StickTogether buffer)	New England Biolabs #M0318L
D-Biotin	Invitrogen #B1595 <i>or equivalent</i>
Unloaded Tn5 protein	Diagenode #C01070010
Tris pH8	<i>available from many suppliers</i>
NaCl	<i>available from many suppliers</i>
NaOH	<i>available from many suppliers</i>
EDTA	<i>available from many suppliers</i>
Glycerol	<i>available from many suppliers</i>
Triton X-100	<i>available from many suppliers</i>
Nuclease-free water	<i>available from many suppliers</i>
HEPES pH 7.3	ThermoFisher #J16924 <i>or equivalent</i>
DMSO (dimethylsulfoxide)	<i>available from many suppliers</i>

List of oligo duplexes and custom oligos:

For oligo sequences, see BLinkBead_oligos.xlsx

2xBiotin-TS5P duplex	2xBio-TS5P-5p-ACA / TS5P-5pRev
Stagger-duplexes	TS5P-3p-S[0-7] / TS5P-3pRev-S[0-7], 8-plex series with staggered lengths
Barcode-duplexes	12nt BC duplexes with 3nt sticky ends, 3 x 96-well plates Segment 1 = 5'Phos-CGA-12ntBC / 12ntRCBC-TTC Segment 2 = 5'Phos-CTA-12ntBC / 12ntRCBC-TCG Segment 3 = 5'Phos-CCA-12ntBC / 12ntRCBC-TAG
ME-LIG-duplex	Ph-shortME / MEREV-LIG-TGG
ME-REV-X	/5Phos/CTGTCTCTTATACACATCT/3InvdT/ <i>recommended but not required: add 3'Inverted-dT to block 3' extension of the ME-REV oligo</i>
ME-B	GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAG

Equipment and supplies:

Pipettors (including multichannels), vortex (tube and plate), mini-centrifuge (tube, strip tube, and plate), Thermomixer(s) with tube and 96-well plate adapters, tube and plate rotator, PCR machine, magnetic racks (tube and plate).

Tubes (1.5mL, 5mL, 200ul PCR strip tubes, 15mL and 50mL falcon tubes), plates, filtered pipet tips, reagent reservoirs

Buffer Recipes:

10x Annealing buffer

Store at room temperature.

To prepare 1x Annealing buffer, dilute 10-fold in NF water, store at room temperature

volume	stock reagent	10x conc.
5 mL	5 M NaCl	500 mM
5 mL	1 M Tris, pH 8.0	100 mM
40 mL	NF water	
50 mL		

5x Streptavidin binding buffer (SBB)

Store at room temperature

* To prepare 1x SBB+T, dilute 5-fold in NF water and add 1:100 vol 10% Triton X-100 (final 0.1%)

volume	stock reagent	5x buffer
30 mL	5 M NaCl	3 M
2.5 mL	1 M Tris, pH 8.0	50 mM
250 µl	500 mM EDTA	2.5 mM
17.25 mL	NF water	
50 mL		

10x WASH buffer

Store at room temperature

* To prepare 1x WASH+T, dilute 10-fold in NF water and add 1:100 vol 10% Triton X-100 (final 0.1%)

* To prepare 1x WASH+biotin, add 125µL 200mM D-Biotin per 50mL 1x WASH+T (final 0.5mM D-Biotin)

To prepare 200mM D-Biotin, dissolve 49mg D-Biotin in 1mL DMSO

volume	stock reagent	10x buffer
3 mL	5 M NaCl	300 mM
5 mL	1 M Tris, pH 8.0	100 mM
42 mL	NF water	
50 mL		

Tn5 dialysis/storage buffer (30% glycerol for loaded beads)

Store at 4°C

Adjust the glycerol to final 50% to use this buffer for Tn5 protein dilution and storage at 20°C

volume	stock reagent	final conc.
15 mL	100% glycerol	30%
2.5 mL	1 M HEPES pH 7.3	50 mM
2 mL	5 M NaCl	200 mM
20 µl	500 mM EDTA	0.2 mM
1 mL	10 % Triton X-100	0.2 %
29.5 mL	NF water	
50 mL	50 mL	

NaOH stripping buffer

* Prepare fresh: 0.15 M NaOH, 0.1% Triton

* Buffers containing 0.1% Triton should be prepared fresh daily. Do not store long term.

Recommended: mix by inversion, as vortexing will generate foam.

Appendix B: Preparation of duplex-oligo stocks

Buffers:

10x Annealing buffer 100mM Tris pH 8, 500mM NaCl

Duplex oligo stocks required for combinatorial barcode ligations:

25μM 2xBiotin-TS5P duplex	2xBio-TS5P-5p-ACA / TS5P-5pRev
25μM Stagger-duplexes	TS5P-3p-S[0-7] / TS5P-3pRev-S[0-7], 8-plex series with staggered lengths
10μM Barcode segment-duplexes	12nt BC duplexes with 3nt sticky ends, 3 x 96-well plates Segment 1 = 5'Phos-CGA-12ntBC / 12ntRCBC-TTC Segment 2 = 5'Phos-CTA-12ntBC / 12ntRCBC-TCG Segment 3 = 5'Phos-CCA-12ntBC / 12ntRCBC-TAG
10μM ME-LIG-duplex	Ph-shortME / MErev-LIG-TGG

Protocol:

1. Prepare an equimolar mixture of the top-strand and bottom-strand oligos in 1x Annealing buffer in a PCR tube. Dilute each individual oligo to the final concentration desired for the duplex-oligo stock.

For example, to prepare 100μl of 25 μM 2xBiotin-TS5P duplex oligos, combine:

*40 μl nuclease-free water
10 μl 10x Annealing buffer
25 μl 100μM 2xBio-TS5P-5p-ACA oligo
25 μl 100μM TS5P-5pRev oligo*

2. Heat and slowly cool the oligos to promote duplex formation:
65°C/5min – slow cool 1°C/minute to 22°C.
3. Store duplex-oligo stocks at -20°C. Dilute in 1x Annealing buffer as needed.